Southern New Hampshire University

CS 255 Business Requirements Document Template

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of the DriverPass system is to help students better understand and prepare for their DMV driving tests by providing access to different online classes, practice tests, and behind-the-wheel driving lessons. DriverPass wants to help students that struggle with their driving tests gain more experience and confidence before taking the actual exam.

- The client for this project is DriverPass, a company owned by Liam that wants to provide better driver training to students who are preparing for their DMV driving tests. Liam sees a need for more opportunities for students to practice and learn before taking their tests. The purpose of the system is to support this goal by providing online learning resources and scheduling for behind-the-wheel driving lessons.

- The system will also help DriverPass manage its business more efficiently. Customers will be able to register for services, choose training packages, and manage their appointments, while employees can keep track of students, drivers, vehicles, and training schedules in one place. This will help the company stay organized while providing the training services that the customers require.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass wants the system to bring its customer service and daily operations together in one place. Customers should be able to create accounts, purchases training packages, and manage their driving lesson appointments online. The system should also connect each appointment with the appropriate driver, vehicle, and time slot to help prevent scheduling conflicts. Employees will use the same system to manage customer information and keep track of the company's services, making it easier to coordinate lessons and keep records organized
- One problem DriverPass wants to address is that students may not get enough practice or preparation before taking their DMV driving tests. The company also needs a better way to manage it daily operations because it has multiple customers, drivers, appointments, and training packages to keep track of. Managing these details separately could make it harder to keep information organized and coordinate driving lessons. Having a central system will help DriverPass keep track of its business activities and make it easier to manage its growing number of customers and training services.
- The system will need serval main components to support DriverPass's services and daily operations. These include customer account management, appointment scheduling, training package management, online learning materials, and employee management. Customer account management will keep students' information organized, while appointment scheduling will help employees coordinate driving lessons with available drivers and vehicles. Training package management will keep track of what customers purchase, and the online learning component will give students access to classes and practice exams. Employee management will

allow staff to access the information they need for their jobs. These components will work together through one “Online System”, allowing DriverPass to manage its services more efficiently and keep its information organized.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The completed DriverPass system should be able to allow customers to create an account, purchase different training packages, schedule/cancel, and modify driving lessons online. It should also match each two-hour lesson with an available driver, vehicle, and time slot to prevent any scheduling conflicts. Students should also be able to access online classes, take practice exams, and view their test progress, including scores and test status. These features will help students prepare for their DMV driving tests and will also at the same time make it easier to manage their training.
- To achieve these goals, the system design must include customer account management, appointment scheduling, training management, online learning and employee management. It also includes role-based access, password resets, and the ability to block employee accounts when needed. The system should record who creates, changes or cancels reservations and allow authorized employees to generate and print activity reports, and let drivers record lesson times and student progress notes. Finally, it should receive DMV updates and notify DriverPass when there’s some sort of new rules, policies, or sample questions become available. Secure cloud hosting, backups and web access from computers and mobile devices will help keep the system reliable and accessible.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- DriverPass should be accessible through a web browser on computers, tablets, and smartphones, allowing customers and employees to use the system from different locations without installing a separate application. Pages should load fast, mainly within a second or two, and actions like logging in, searching for available appointments times and submitting information should also respond quickly. Also, information should be updated promptly when changes are made. For example, when a driving lesson is scheduled or canceled, the available time slots should reflect that change in a timely manner.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- DriverPass should run through a web browser and be accessible from computers and mobile devices, allowing customers and employees to access the system from different locations. The back end will need a database to store customer accounts, training packages, appointments, driver and vehicle information, employee records, and student progress notes. Secure cloud hosting will also support access to the database from different locations while keeping business records organized in the one central location

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Each customer and employee should have a separate account so that the system can identify who is accessing or changing information with case sensitive account names for variety. The system should accurately record lesson reservations, cancellations, and modifications, including who made the change and when it occurred. It should also keep track of driving appointments, including the assigned driver, vehicle, and scheduled time, to help prevent scheduling conflicts. If something goes wrong, the system should allow authorized employees to print activity reports to review what happened and determine who made the change. Also, the system should keep accurate student records, including lesson times and instructor notes, so employees can easily review student progress when needed.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- DriverPass should allow authorized employees to add, remove and modify user accounts without changing the program code. The system should also be designed so it can be updated to support newer web browsers and operating systems as technology changes. These updates should be tested before being put into use to make sure that the customers and employees can still access the system and that features like scheduling, training materials, and customer records continue to work properly. The IT admin should have administrative access to manage user accounts like resetting passwords, and adding or removing access when needed since employees may change roles or leave the company, and users may need help regaining access to their account.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- DriverPass users should log in with a unique username or email address and password so the system can identify who is accessing the system. To protect customer information and business records, the system should use encryption when data is exchanged between users' devices and the server. As an additional security measure, the system should temporarily lock an account or

require some sort of additional verification after several failed login attempts to help prevent against brut force hacking. Users who forget their passwords should be able to reset them through a secure process that verifies their identity before allowing them to create a new password.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow customers to create accounts and register for DriverPass services.
- The system shall allow customers to purchase training packages that include the services offered by DrivePass like online classes, and behind the wheel driving lessons.
- The system shall allow customers to schedule, cancel, and modify driving lesson appointments online.
- The system shall match each driving lesson with an available driver, vehicle, and time slot to prevent scheduling conflicts.
- The system shall allow the secretary to schedule, cancel, and modify appointments on behalf of customers who call or visit DriverPass.
- The system shall allow students to access online classes and practice tests to prepare for their driving exams.
- The system shall track student test progress, including test names, scores, time, taken, and status, such as not taken, in progress, failed or passed.
- The system shall allow drivers to record lesson times and enter notes about each student's driving progress.
- The system shall allow authorized employees to generate and print reports about reservations, cancellations, and appointment changes.
- The system shall record who created, canceled, or modified an appointment and when the change occurred.
- The system shall allow the IT administrator to manage user access, reset passwords, and disable accounts when necessary.
- The system shall notify DriverPass when the new DMV rules, policies, or training materials become available.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The needs of the interface are to provide an easy way for customers and employees to access the features they need because each user has different responsibilities within DriverPass. Customers need to register for accounts, purchase training packages, schedule or modify driving lessons, and access online classes and practice tests. The different users of the interface include customers, the secretary, drivers, other authorized employees, and the IT administrator. Customers need to manage their training and appointments, the secretary needs to assist customers withing scheduling, drivers need to view their schedules and record lesson times and

student progress notes, authorized employees need to generate and print reports, and the IT administrator needs to manage user accounts and access permissions. Users will interact with the system through a web browser on computers or mobile devices, allowing them to access DriverPass from different locations.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- The design assumes that customers and employees will have access to a computer or mobile device with an internet connection to use the web-based DriverPass system. It also assumes that users will have basic computer skills to navigate the interface, create an account, schedule appoints, and access training materials. Since the system will store customer information. Appointment records, and student progress, it is assumed that DriverPass will have a database and a secure online hosting to keep this information available to authorized users. The design also assumes that DriverPass employees will keep customer records, driver schedules, and training information up to date so the system can provide accurate information.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- One limitation of the DriverPass system is that it will rely on the accuracy of the information entered by employees. If a driver's availability. Vehicle information, or appointment details are not updated correctly, the system may display incorrect availability or allow scheduling conflicts. The system will also depend on DriverPass receiving updated DMV rules and training materials to keep its online classes and practice tests current. If new information is delayed, students may not have access to the latest material when preparing for their driving tests.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

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DriverPass Transcript Project

